ENGR 2405 – ELECTRICAL CIRCUITS I

CLASS INFORMATION

Synonym: 50115
Section: 001
Location: HLC1000/Rm2106 – lecture and lab
Class Times: Mon/Wed 10:00A - 2:10P

INSTRUCTOR SPECIFIC INFORMATION

Name: Purna Murthy
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Phone Numbers: 512-223-7931
Office Hours: Mon/Wed 2:10P - 3:10P
Tue/Thu 12:00P - 1:30P
or by appointment

COURSE DESCRIPTION

Credit Hours: 4
Classroom Contact Hours per week: 3
Laboratory Contact Hours per week: 3

Principles of electrical circuits and systems. Basic circuit elements (resistance, inductance, mutual inductance, capacitance, independent and dependent controlled voltage, and current sources). Topology of electrical networks; Kirchhoff's laws; node and mesh analysis; DC circuit analysis; operational amplifiers; transient and sinusoidal steady-state analysis; AC circuit analysis; first- and second-order circuits; Bode plots; and use of computer simulation software to solve circuit problems. Laboratory experiments supporting theoretical principles presented in lectures involving DC and AC circuit theory, network theorems, time, and frequency domain circuit analysis. Introduction to principles and operation of basic laboratory equipment; laboratory report preparation.

COURSE RATIONALE/OBJECTIVES

This course is a comprehensive introduction to the fundamentals of electric circuits intended. This course is intended to develop student skills in analyzing and solving a variety of linear DC and AC circuits through calculations, computer simulations, and lab experiments.

COURSE PREREQUISITES:

- MATH 2414 (Calculus II) and PHYS 2426 (Engineering Physics II), or equivalents
- Co-requisite: MATH 2420 (Differential Equations), or previous completion

Required Texts/Materials

- Alexander, Charles / Sadiku, Matthew, Fundamentals of Electric Circuits, 7th edition
- Computer with SciLab and LTSpice circuit simulator (free software downloads)
- Scientific calculator (graphing not required)

Instructional Methodology

This is a lecture with a lab course which includes time for class discussions, demonstrations, student projects, labs and/or activities guided by the instructor.

Student Technology Support

Austin Community College provides free, secure drive-up Wi-Fi to students and employees in the parking lots of all campus locations. Wi-Fi can be accessed seven days a week, 7 am to 11 pm. Additional details are available at **Student Technology Services**.

Student Technology Services offers phone, live-chat, and email-based technical support for students and can provide support on topics such as password resets, accessing or using Blackboard, access to technology, etc. To view hours of operation and ways to request support, visit **Student Technology Services.**

Copyrighted Materials

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Student Learning Outcomes

Course-Level Student Learning Outcomes:

Upon successful completion of this course, the student will be able to:

- Explain basic electrical concepts, including electric charge, current, electrical potential, electrical power, and energy
- Apply concepts of electric network topology: nodes, branches, and loops to solve circuit problems, including the use of computer simulation.
- Analyze circuits with ideal, independent, and controlled voltage and current sources.
- Apply Kirchhoff's voltage and current laws to the analysis of electric circuits.
- Explain the relationship of voltage and current in resistors, capacitors, and inductors.
- Derive and solve the governing differential equations for a time-domain first-order and second-order circuit, including singularity function source models.
- Determine the Thévenin or Norton equivalent of a given network that may include passive devices, dependent sources, and independent sources in combination.

- Analyze first and second order AC and DC circuits for steady-state and transient response in the time domain and frequency domain.
- Derive relations for and calculate the gain and input impedance of a given operational amplifier circuit for both DC and frequency domain AC circuits using an ideal operational amplifier model.
- Apply computer mathematical and simulation programs to solve circuit problems.
- Prepare laboratory reports that clearly communicate experimental information in a logical and scientific manner.
- Conduct basic laboratory experiments involving electrical circuits using laboratory test equipment such as multimeters, power supplies, signal generators, and oscilloscopes.
- Predict and measure the transient and sinusoidal steady-state responses of simple RC, RL, and RLC circuits.
- Predict the behavior and make measurements of simple operational-amplifier circuits.
- Relate physical observations and measurements involving electrical circuits to theoretical principles.
- Evaluate the accuracy of physical measurements and the potential sources of error in the measurements.

General Education Competencies:

Upon completion of this course, students will demonstrate competence in:

Communication Skills
Develop, interpret, and express ideas and information through written, oral and visual communication that is adapted to purpose, structure, audience, and medium.

Critical Thinking Skills
Gather, analyze, synthesize, evaluate and apply information for the purposes of innovation, inquiry, and creative thinking.

Empirical and Quantitative Skills
Apply mathematical, logical and scientific principles and methods through the manipulation and analysis of numerical data or observable facts resulting in informed conclusions.

Teamwork
Consider different points of view to work collaboratively and effectively in pursuit of a shared purpose or goal.

COURSE POLICIES

GRADING SYSTEM

Grades will be recorded on Blackboard and course grade determined by the following:

- A: 90–100
- B: 80 – 89
- C: 70 – 79
- D: 60 – 69
- F: 59 or lower

Lecture Portion (75%)

Tests (90%)

- There will be 3 unit exams each worth 30% - **no exam may be dropped**

Homework (10%)

- Homework will be assigned for the course generally by module.

Lab Portion (25%)

- **Lab Reports** (70%): The top 11 (out of 12) reports will be used
- **Lab Exam** (30%): There will be a lab practicum exam

Note: Personal electronic devices (including cell phones) will be turned off during class and left in your backpack or other container. Scientific calculators are an exception to this rule.

A student must earn a grade of "C" or better in both the laboratory and lecture portions of the course in order to earn a grade of "C" or better in the course. If they do not earn a grade of "C" or better in each of these portions, then their grade for the course will be, at most, a "D". The grade in each portion, either lecture or laboratory, of the course will be as outlined in the syllabus and the determination of the grades will be as stated in the syllabus.

HOMEWORK POLICY

No late work is accepted except for extreme circumstances.

EXAM AND TEST POLICIES

All exams will be closed book and closed notes (a formula sheet may be provided). No cell phones or computers will be required and only blank "scratch" paper and a scientific calculator will be allowed to be used during exams.

Attendance/Class Participation

Regular and punctual class and laboratory attendance is expected of all students. If attendance or compliance with other course policies is unsatisfactory, the instructor may withdraw students from the class. **After 3 unexcused absences of scheduled class meetings the instructor may drop the student from class.**

During class, all laptops, tablets, cell phones, ear buds, etc must be removed and put away into backpacks, purses, etc. If you are using any of these items during class, you will be asked to leave for the remainder of the class. The only exception will be laptops/tablets for express purpose of taking notes.

In the event the college or campus closes due to unforeseen circumstances (for example, severe weather or other emergency), the student is responsible for communicating with their professor during the closure and completing any assignments or other activities designated by their professor as a result of class sessions being missed.

Withdrawal Policy

It is the responsibility of each student to ensure that his or her name is removed from the rolls should they decide to withdraw from the class. The instructor does, however, reserve the right to drop a student should he or she feel it is necessary. **If a student decides to withdraw, he or she should also verify that the withdrawal is recorded <u>before</u> the Final Withdrawal Date, found on the ACC Academic Calendar page: https://www.austincc.edu/students/registration/important-dates**. The student is also strongly encouraged to keep any paperwork in case a problem arises.

Students are responsible for understanding the impact that withdrawal from a course may have on their financial aid, veterans' benefits, and international student status. Per state law, students enrolling for the first time in Fall 2007 or later at any public Texas college or university may not withdraw (receive a W) from more than six courses during their undergraduate college education. Some exemptions for good cause could allow a student to withdraw from a course without having it count toward this limit. Students are strongly encouraged to meet with an advisor when making decisions about course selection, course loads, and course withdrawals.

Missed Exam/Late Work Policies

Arrangements for taking an exam early for any missed exam should be made in advance if at all possible. No makeup exams will be given except for extreme circumstances.

INCOMPLETES

A grade of incomplete should be reserved only for extreme cases meeting the following criteria.

The student has had a documented life event beyond their control that will prevent them from completing the semester on time.

The student is in good standing (Grade of 'C' or better at the time of the life event from 1)

The student has completed most of the material in the course.

Before assigning a grade of incomplete, the instructor and the student must agree to a plan of action that includes a specific list of tasks to be completed by the student with a timeline of completion. This plan needs to be approved by the department chair (or designee). Incompletes must be resolved before the final withdrawal date of the following semester.

Students may request an Incomplete from their faculty member if they believe circumstances warrant. The faculty member will determine whether the Incomplete is appropriate to award or not. The following processes must be followed when awarding a student an I grade.

1. Prior to the end of the semester in which the "I" is to be awarded, the student must meet with the instructor to determine a plan of action that identifies all of the assignments and exams that must be completed prior to the deadline date. This meeting can occur virtually or in person. The instructor should complete the Report of Incomplete Grade form with the plan of action and send it to the department chair (or designee) to be approved.
2. Once approved, the faculty member will complete the form, including all requirements to complete the course and the due date, sign (by typing in name) and then email it to the student. The student will then complete his/her section, sign (by typing in name), and return the completed form to the faculty member to complete the agreement. A copy of the fully completed form can then be emailed by the faculty member to the student and the department chair for each grade of Incomplete that the faculty member submits at the end of the semester.
3. The student must complete all remaining work by the date specified on the form above. This date is determined by the instructor in collaboration with the student, but it may not be later than the final withdrawal deadline in the subsequent long semester.
4. Students will retain access to the course Blackboard or other LMS named here through the subsequent semester in order to submit work and complete the course. Students will be able to log on to Blackboard or other LMS named here and have access to the course section materials, assignments, and grades from the course and semester in which the Incomplete was awarded.
5. When the student completes the required work by the Incomplete deadline, the instructor will submit an electronic Grade Change Form to change the student's performance grade from an "I" to the earned grade of A, B, C, D, or F.

If an Incomplete is not resolved by the deadline, the grade automatically converts to an "F." Approval to carry an Incomplete for longer than the following semester or session deadline is not frequently granted.

COLLEGE POLICIES AND STUDENT SUPPORT SERVICES

See separate link in Blackboard

COURSE OUTLINE/COURSE CALENDAR

Please note that schedule changes may occur during the semester. Any changes will be announced in class and posted as a Blackboard Announcement (or other resource faculty is using to communicate).

	06/01/2026	**Summer 2026**	
Week	**Dates**	**Mon**	**Wed**
W01	06/01/2026 - 06/05/2026	FDM / CH01-02 / LAB01	CH03 / CH04a
W02	06/08/2026 - 06/12/2026	CH04b / LAB04	**Unit01 Exam**
W03	06/15/2026 - 06/19/2026	CH05 / CH06a	LAB05
W04	06/22/2026 - 06/26/2026	CH06b / CH07a	LAB06
W05	06/29/2026 - 07/03/2026	CH07b / CH08	LAB07
W06	07/06/2026 - 07/10/2026	LAB08	**Unit02 Exam**
W07	07/13/2026 - 07/17/2026	CH09-10 / CH11-12a	LAB09
W08	07/20/2026 - 07/24/2026	CH11-12b / CH14a	LAB10
W09	07/27/2026 - 07/31/2026	CH14b / LAB11	LAB12
W10	08/03/2026 - 08/07/2026	**Practicum (1h/person)**	**Unit03 Exam**

Note:
LAB02 and LAB03 are take-home labs